

UNIT 1

Activity: Top 5 Cities by Average Revenue per Category

CO Mapped: CO1 –Use Big Data and its Business Implications.

Type: Practical + Analytical Activity

Problem Statement

In large-scale e-commerce systems, businesses operate across multiple cities and product categories. Each transaction generates revenue based on the quantity of items sold and their price. To analyze business performance, it is important to determine which cities generate the highest average revenue within each product category. Given a large sales dataset containing transaction details such as Category, City, Quantity, and Price, develop a MapReduce program to:

1. Calculate the average revenue for each (Category, City) pair.
2. Identify the Top 5 cities with the highest average revenue in each category.

Sample Input Dataset

`e-commerce_transaction.csv`

```
TransactionID,UserID,ProductID,Category,City,Quantity,Price,PaymentMode,Date
T1001,U101,P501,Electronics,Mumbai,1,25000,CreditCard,2026-02-14
T1002,U102,P502,Fashion,Delhi,3,1500,UPI,2026-02-15
T1003,U103,P503,Grocery,Bangalore,5,200,Cash,2026-02-16
T1004,U104,P504,Electronics,Chennai,2,18000,DebitCard,2026-02-17
T1005,U105,P505,Fashion,Kolkata,4,2200,UPI,2026-02-18
T1006,U106,P506,Electronics,Hyderabad,1,32000,CreditCard,2026-02-18
```

Objective

This activity focuses on implementing a multi-stage MapReduce solution to compute the Top 5 Cities by Average Revenue per Category from a sales dataset. Through this activity, student able to compute distributed averages, apply reducer-side sorting, and extract Top-K results using Hadoop MapReduce..

Tasks to Perform

Task 1: Implement Revenue Calculation using Mapper

This task establishes revenue extraction logic. Through this task, student able to parse structured dataset and compute transaction revenue..

- Read input dataset from HDFS.
- Extract Category, City, Quantity, and Price fields.
- Compute revenue as $\text{Quantity} \times \text{Price}$.
- Emit key as “Category|City”.
- Emit revenue as value.

Task 2: Implement Average Revenue Calculation

- Receive all revenue values for each (Category|City).
- Compute total revenue and count of transactions.
- Calculate average revenue inside Reducer.
- Output Category|City $\rightarrow$ AverageRevenue.

Task 3: Implement Category-wise Grouping for Ranking

- Read output of Job 1.
- Separate Category and City from key.
- Emit key as Category.
- Emit value as City:AverageRevenue.
- Ensure correct grouping by Category.

Task 4: Extract Top 5 Cities per Category

- Receive all cities belonging to same category in Reducer.
- Store values in suitable list structure.

- Sort list in descending order of average revenue.
- Output only Top 5 cities per category.
- Format final output clearly.
- Capture screenshots of successful execution

Task 5: Written Summary Document

- Prepare a 500-word report
- Explain Mapper and Reducer logic
- Describe preprocessing steps
- Provide sample intermediate and final output
- Discuss challenges faced

What Students Must Submit

Students must submit:

- Source code for both MapReduce jobs
- Sample input dataset
- Execution screenshots
- Final output file
- Short explanation describing why Combiner was not used
- Brief comparison of performance with and without Combiner
- Single compressed folder or GitHub repository link

Learning Outcome

Upon completion of this activity, students will be able to:

- Implement MapReduce aggregation
- Understand shuffle and reducer workload
- Compute distributed averages correctly
- Perform group-based ranking

Rubric

Task	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Poor (1 Mark)	No Attempt (0 Mark)
Task 1: Revenue Extraction using Mapper	Correctly parses dataset, computes revenue (Quantity $\times$ Price), handles invalid records, and emits accurate key-value pairs.	Correct parsing and revenue calculation with minor formatting issues.	Basic parsing works but revenue calculation or key formatting has errors.	Mapper exists but produces incorrect or inconsistent output.	Mapper not implemented.
Task 2: Average Revenue Calculation in Reducer	Accurately computes total revenue, transaction count, and average for each (Category, City) entirely inside Reducer.	Average calculation mostly correct with minor logical issues.	Partial aggregation implemented but average incorrect or inconsistent.	Reducer logic incomplete or produces incorrect results.	Reducer not implemented.
Task 3: Category-wise Grouping for Ranking	Correctly reorganizes data by Category and prepares proper input for Top-K extraction.	Grouping logic works with minor formatting issues.	Grouping partially correct but some misclassification exists.	Grouping logic flawed or inconsistent.	Grouping step not implemented.
Task 4: Top 5 Extraction per Category	Correctly sorts cities in descending order and outputs accurate Top 5 per category with proper formatting.	Top 5 extraction works with minor sorting or formatting issues.	Sorting implemented but Top 5 logic partially incorrect.	Ranking logic exists but fails to produce correct results.	Top 5 extraction not implemented.
Task 5: Written Summary Document (500 Words)	Well-structured 500-word report clearly explaining Mapper logic, Reducer logic, preprocessing steps, intermediate output, final output, and challenges faced with proper technical clarity.	Report covers most required points with minor clarity or structural issues.	Report addresses some required points but lacks depth or completeness.	Report submitted but incomplete, unclear, or below required length.	No written summary submitted.